

- Description

This module manages a smart subway station passenger flow monitoring system, ensuring that:

1. **The number of passengers entering and exiting the station is continuously monitored.**
2. **If the station reaches maximum capacity, turnstiles restrict new entries until the congestion level decreases.**
3. **If the station is underutilized, more turnstiles open to improve passenger flow efficiency.**
4. **A log of all passenger counts, congestion alerts, and turnstile adjustments is maintained for real-time management.**

This system helps prevent overcrowding, optimize station efficiency, and enhance passenger safety.

- Specification

module SmartSubwayStationMonitoring

var currentPassengers, maxCapacity, openTurnstiles: integer

var log: sequence of string

process enterStation(numPassengers: integer)

functional scenario form:

currentPassengers + numPassengers <= maxCapacity && currentPassengers = currentPassengers + numPassengers, adjustTurnstiles(), log = log $\oplus$ ("Entry: " + numPassengers + " passengers | New Count: " + currentPassengers + " | Open Turnstiles: " + openTurnstiles), Passengers entered, turnstile count adjusted.

||

currentPassengers + numPassengers > maxCapacity && log = log $\oplus$ ("Overcrowding Alert: Entry Restricted."), Overcrowding alert triggered, entry restricted.

end process

process exitStation(numPassengers: integer)

functional scenario form:

currentPassengers = max(0, currentPassengers - numPassengers), adjustTurnstiles(), log = log $\oplus$ ("Exit: " + numPassengers + " passengers | New Count: " + currentPassengers + " | Open Turnstiles: " + openTurnstiles) Passengers exited, turnstile count adjusted.

end process

end module

- Java Code

```
package code;
import java.util.*;
class SubwayStation {
    private int maxCapacity;
```

```

private int currentPassengers;
private int openTurnstiles;
private List<String> log;
public SubwayStation(int maxCapacity, int initialPassengers) {
    this.maxCapacity = maxCapacity;
    this.currentPassengers = initialPassengers;
    this.openTurnstiles = 5;
    this.log = new ArrayList<>();
}
public void enterStation(int numPassengers) {
    if (currentPassengers + numPassengers > maxCapacity) {
        System.out.println("WARNING: Station Overcrowded! Entry Restricted.");
        log.add("Overcrowding Alert: Entry Restricted.");
        return;
    }
    currentPassengers += numPassengers;
    adjustTurnstiles();
    System.out.printf("Passengers Entered: %d | Current Count: %d | Open Turnstiles: %d\n", numPassengers, currentPassengers, openTurnstiles);
    log.add(String.format("Entry: %d passengers | New Count: %d | Open Turnstiles: %d", numPassengers, currentPassengers, openTurnstiles));
}
public void exitStation(int numPassengers) {
    currentPassengers = Math.max(0, currentPassengers - numPassengers);
    adjustTurnstiles();
    System.out.printf("Passengers Exited: %d | Current Count: %d | Open Turnstiles: %d\n", numPassengers, currentPassengers, openTurnstiles);
    log.add(String.format("Exit: %d passengers | New Count: %d | Open Turnstiles: %d", numPassengers, currentPassengers, openTurnstiles));
}
private void adjustTurnstiles() {
    if (currentPassengers > maxCapacity * 0.8) {
        openTurnstiles = 3;
    } else if (currentPassengers < maxCapacity * 0.3) {
        openTurnstiles = 8;
    } else {
        openTurnstiles = 5;
    }
}
}

```

```

    public void printLog() {
        System.out.println("\nSubway Station Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c43_SubwayPassengerFlowSystem {
    public static void main(String[] args) {
        SubwayStation station = new SubwayStation(1000, 400);
        // testcase-VT:
        station.enterStation(-200);
        station.enterStation(-270);
        station.enterStation(-135);
        station.enterStation(-474);
        station.enterStation(320);
        station.enterStation(-259);
        station.enterStation(-471);
        station.enterStation(172);
        station.enterStation(210);
        station.enterStation(402);
        station.enterStation(262);
        station.enterStation(-243);
        station.enterStation(-391);
        station.enterStation(-420);
        station.enterStation(-225);
        station.enterStation(55);
        station.enterStation(238);
        station.enterStation(-246);
        station.enterStation(261);
        station.enterStation(-121);
        station.enterStation(-387);
        station.enterStation(276);
        station.enterStation(110);
        station.enterStation(153);
        station.enterStation(-141);
        station.enterStation(-440);
        station.enterStation(47);
        station.enterStation(142);
        station.enterStation(-129);
        station.enterStation(-410);
        station.enterStation(373);
        station.enterStation(-171);
        station.enterStation(194);
    }
}

```

```
station.enterStation(358);
station.enterStation(-492);
station.enterStation(-489);
station.enterStation(84);
station.enterStation(204);
station.enterStation(70);
station.enterStation(-322);
station.enterStation(334);
station.enterStation(-476);
station.enterStation(-183);
station.enterStation(327);
station.enterStation(-100);
station.enterStation(-215);
station.enterStation(-320);
station.enterStation(231);
station.enterStation(-102);
station.enterStation(-213);
station.exitStation(-491);
station.exitStation(75);
station.exitStation(-147);
station.exitStation(216);
station.exitStation(-343);
station.exitStation(422);
station.exitStation(262);
station.exitStation(161);
station.exitStation(-94);
station.exitStation(45);
station.exitStation(354);
station.exitStation(-473);
station.exitStation(181);
station.exitStation(-356);
station.exitStation(-75);
station.exitStation(463);
station.exitStation(-219);
station.exitStation(246);
station.exitStation(119);
station.exitStation(262);
station.exitStation(251);
station.exitStation(-168);
station.exitStation(450);
station.exitStation(276);
station.exitStation(-491);
station.exitStation(-144);
station.exitStation(-447);
```

```
station.exitStation(454);
station.exitStation(-43);
station.exitStation(197);
station.exitStation(-210);
station.exitStation(248);
station.exitStation(337);
station.exitStation(-207);
station.exitStation(-123);
station.exitStation(89);
station.exitStation(-326);
station.exitStation(-238);
station.exitStation(-16);
station.exitStation(482);
station.exitStation(299);
station.exitStation(-39);
station.exitStation(419);
station.exitStation(-257);
station.exitStation(-468);
station.exitStation(-226);
station.exitStation(-499);
station.exitStation(130);
station.exitStation(19);
station.exitStation(63);

// station.enterStation(1835925503);
// station.enterStation(-1);
// station.enterStation(-16711681);
// station.enterStation(-16711681);
// station.enterStation(-16711901);
// station.enterStation(-16711901);
// station.enterStation(109);
// station.enterStation(111);
// station.enterStation(139);
// station.enterStation(109);
// station.enterStation(28013);
// station.enterStation(28013);
// station.enterStation(44141);
// station.enterStation(46701);
// station.enterStation(11937389);
// station.enterStation(2434669);
// station.enterStation(237);
// station.enterStation(28141);
// station.enterStation(28013);
// station.enterStation(28013);
```

```
//      station.enterStation(7171437);
//      station.enterStation(1);
//      station.enterStation(0);
//      station.enterStation(0);
//      station.enterStation(0);
//      station.enterStation(0);
//      station.enterStation(109);
//      station.enterStation(28013);
//      station.enterStation(109);
//      station.enterStation(27904);
//      station.enterStation(512);
//      station.enterStation(1828716544);
//      station.enterStation(1828716560);
//      station.enterStation(-1828716560);
//      station.enterStation(-1828716560);
//      station.enterStation(-1828716798);
//      station.enterStation(109);
//      station.enterStation(28098);
//      station.enterStation(27965);
//      station.enterStation(15725);
//      station.enterStation(61);
//      station.enterStation(7143424);
//      station.enterStation(1828716682);
//      station.enterStation(1828716682);
//      station.enterStation(1828716682);
//      station.enterStation(1828716682);
//      station.enterStation(1);
//      station.enterStation(9);
//      station.enterStation(6);
//      station.enterStation(1662);
//      station.enterStation(1662);
//      station.enterStation(59);
//      station.enterStation(38);
//      station.enterStation(10022);
//      station.enterStation(14118);
//      station.enterStation(13350);
//      station.enterStation(6253);
//      station.enterStation(24);
//      station.enterStation(1);
//      station.enterStation(129);
//      station.enterStation(125);
//      station.enterStation(64);
//      station.enterStation(1128481603);
//      station.enterStation(1128481603);
```

```
//      station.enterStation(1128481603);
//      station.enterStation(91);
//      station.enterStation(1543503871);
//      station.enterStation(1543503871);
//      station.enterStation(1543503871);
//      station.enterStation(1543503871);
//      station.enterStation(109);
//      station.enterStation(0);
//      station.enterStation(0);
//      station.enterStation(0);
//      station.enterStation(0);
//      station.enterStation(28013);
//      station.enterStation(7171376);
//      station.enterStation(1835875632);
//      station.enterStation(1835872317);
//      station.enterStation(28013);
//      station.enterStation(109);
//      station.enterStation(125);
//      station.enterStation(109);
//      station.enterStation(16843009);
//      station.enterStation(16843009);
//      station.enterStation(27954);
//      station.enterStation(27954);
//      station.enterStation(1835872818);
//      station.enterStation(1835872562);
//      station.enterStation(1835872562);
//      station.enterStation(28123);
//      station.enterStation(14380507);
//      station.enterStation(0);
//      station.enterStation(0);
//      station.enterStation(111);
//      station.enterStation(111);
//      station.enterStation(1870166136);
//      station.enterStation(1870166136);
//      station.enterStation(1870166136);
//      station.enterStation(28013);
//
//      // testcase-Z3:
//      station.enterStation(-200);
//      station.enterStation(-270);
//      station.enterStation(-135);
//      station.enterStation(-474);
//      station.enterStation(320);
//      station.enterStation(-259);
```

```
// station.enterStation(-471);
// station.enterStation(172);
// station.enterStation(210);
// station.enterStation(402);
// station.enterStation(262);
// station.enterStation(-243);
// station.enterStation(-391);
// station.enterStation(-420);
// station.enterStation(-225);
// station.enterStation(55);
// station.enterStation(238);
// station.enterStation(-246);
// station.enterStation(261);
// station.enterStation(-121);
// station.enterStation(-387);
// station.enterStation(276);
// station.enterStation(110);
// station.enterStation(153);
// station.enterStation(-141);
// station.enterStation(-440);
// station.enterStation(47);
// station.enterStation(142);
// station.enterStation(-129);
// station.enterStation(-410);
// station.enterStation(373);
// station.enterStation(-171);
// station.enterStation(194);
// station.enterStation(358);
// station.enterStation(-492);
// station.enterStation(-489);
// station.enterStation(84);
// station.enterStation(204);
// station.enterStation(70);
// station.enterStation(-322);
// station.enterStation(334);
// station.enterStation(-476);
// station.enterStation(-183);
// station.enterStation(327);
// station.enterStation(-100);
// station.enterStation(-215);
// station.enterStation(-320);
// station.enterStation(231);
// station.enterStation(-102);
// station.enterStation(-213);
```

```
//      station.exitStation(-491);
//      station.exitStation(75);
//      station.exitStation(-147);
//      station.exitStation(216);
//      station.exitStation(-343);
//      station.exitStation(422);
//      station.exitStation(262);
//      station.exitStation(161);
//      station.exitStation(-94);
//      station.exitStation(45);
//      station.exitStation(354);
//      station.exitStation(-473);
//      station.exitStation(181);
//      station.exitStation(-356);
//      station.exitStation(-75);
//      station.exitStation(463);
//      station.exitStation(-219);
//      station.exitStation(246);
//      station.exitStation(119);
//      station.exitStation(262);
//      station.exitStation(251);
//      station.exitStation(-168);
//      station.exitStation(450);
//      station.exitStation(276);
//      station.exitStation(-491);
//      station.exitStation(-144);
//      station.exitStation(-447);
//      station.exitStation(454);
//      station.exitStation(-43);
//      station.exitStation(197);
//      station.exitStation(-210);
//      station.exitStation(248);
//      station.exitStation(337);
//      station.exitStation(-207);
//      station.exitStation(-123);
//      station.exitStation(89);
//      station.exitStation(-326);
//      station.exitStation(-238);
//      station.exitStation(-16);
//      station.exitStation(482);
//      station.exitStation(299);
//      station.exitStation(-39);
//      station.exitStation(419);
//      station.exitStation(-257);
```

```
//      station.exitStation(-468);
//      station.exitStation(-226);
//      station.exitStation(-499);
//      station.exitStation(130);
//      station.exitStation(19);
//      station.exitStation(63);

// testcase-UVT:
station.enterStation(-200);
station.enterStation(-270);
station.enterStation(-135);
station.enterStation(-474);
station.enterStation(320);
station.enterStation(-259);
station.enterStation(-471);
station.enterStation(172);
station.enterStation(210);
station.enterStation(402);
station.enterStation(262);
station.enterStation(-243);
station.enterStation(-391);
station.enterStation(-420);
station.enterStation(-225);
station.enterStation(55);
station.enterStation(238);
station.enterStation(-246);
station.enterStation(261);
station.enterStation(-121);
station.enterStation(-387);
station.enterStation(276);
station.enterStation(110);
station.enterStation(153);
station.enterStation(-141);
station.enterStation(-440);
station.enterStation(47);
station.enterStation(142);
station.enterStation(-129);
station.enterStation(-410);
station.enterStation(373);
station.enterStation(-171);
station.enterStation(194);
station.enterStation(358);
station.enterStation(-492);
station.enterStation(-489);
```

```
station.enterStation(84);
station.enterStation(204);
station.enterStation(70);
station.enterStation(-322);
station.enterStation(334);
station.enterStation(-476);
station.enterStation(-183);
station.enterStation(327);
station.enterStation(-100);
station.enterStation(-215);
station.enterStation(-320);
station.enterStation(231);
station.enterStation(-102);
station.enterStation(-213);
station.exitStation(-491);
station.exitStation(75);
station.exitStation(-147);
station.exitStation(216);
station.exitStation(-343);
station.exitStation(422);
station.exitStation(262);
station.exitStation(161);
station.exitStation(-94);
station.exitStation(45);
station.exitStation(354);
station.exitStation(-473);
station.exitStation(181);
station.exitStation(-356);
station.exitStation(-75);
station.exitStation(463);
station.exitStation(-219);
station.exitStation(246);
station.exitStation(119);
station.exitStation(262);
station.exitStation(251);
station.exitStation(-168);
station.exitStation(450);
station.exitStation(276);
station.exitStation(-491);
station.exitStation(-144);
station.exitStation(-447);
station.exitStation(454);
station.exitStation(-43);
station.exitStation(197);
```

```

        station.exitStation(-210);
        station.exitStation(248);
        station.exitStation(337);
        station.exitStation(-207);
        station.exitStation(-123);
        station.exitStation(89);
        station.exitStation(-326);
        station.exitStation(-238);
        station.exitStation(-16);
        station.exitStation(482);
        station.exitStation(299);
        station.exitStation(-39);
        station.exitStation(419);
        station.exitStation(-257);
        station.exitStation(-468);
        station.exitStation(-226);
        station.exitStation(-499);
        station.exitStation(130);
        station.exitStation(19);
        station.exitStation(63);

//      station.printLog();
    }
}

```

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c43_SubwayPassengerFlowSystem.jav	87.5 %	7	1	8
> SubwayStation	87.5 %	7	1	8
> c43_SubwayPassengerFlowSystem		0	0	0

3. Vibration testing instruction coverage: 97.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.4 %	500	20,523	21,023
▼ src	2.4 %	500	20,523	21,023
▼ code	2.4 %	500	20,523	21,023
▼ c43_SubwayPassengerFlowSystem.jav	97.7 %	500	12	512
> SubwayStation	95.5 %	191	9	200
> c43_SubwayPassengerFlowSystem	99.0 %	309	3	312

4. Fuzz testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c43_SubwayPassengerFlowSystem.jav	87.5 %	7	1	8
> SubwayStation	87.5 %	7	1	8
> c43_SubwayPassengerFlowSystem		0	0	0

5. Fuzz testing instruction coverage: 87.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	2.1 %	446	20,588	21,034
✓ src	2.1 %	446	20,588	21,034
✓ code	2.1 %	446	20,588	21,034
✓ c43_SubwayPassengerFlowSystem.java	87.1 %	446	66	512
> SubwayStation	68.5 %	137	63	200
> c43_SubwayPassengerFlowSystem	99.0 %	309	3	312

6. Z3 branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	834	840
✓ src	0.7 %	6	834	840
✓ code	0.7 %	6	834	840
✓ c43_SubwayPassengerFlowSystem.java	75.0 %	6	2	8
> SubwayStation	75.0 %	6	2	8
> c43_SubwayPassengerFlowSystem		0	0	0

7. Z3 instruction coverage: 97.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	3.5 %	788	21,888	22,676
✓ src	3.5 %	788	21,888	22,676
✓ code	3.5 %	788	21,888	22,676
✓ c43_SubwayPassengerFlowSystem.java	97.3 %	788	22	810
> SubwayStation	90.5 %	181	19	200
> c43_SubwayPassengerFlowSystem	99.5 %	607	3	610

8. Unimproved V-method branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	834	840
✓ src	0.7 %	6	834	840
✓ code	0.7 %	6	834	840
✓ c43_SubwayPassengerFlowSystem.java	75.0 %	6	2	8
> SubwayStation	75.0 %	6	2	8
> c43_SubwayPassengerFlowSystem		0	0	0

9. Unimproved V-method instruction coverage: 97.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	3.5 %	788	21,888	22,676
✓ src	3.5 %	788	21,888	22,676
✓ code	3.5 %	788	21,888	22,676
✓ c43_SubwayPassengerFlowSystem.java	97.3 %	788	22	810
> SubwayStation	90.5 %	181	19	200
> c43_SubwayPassengerFlowSystem	99.5 %	607	3	610